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### **Q1. Hospital Patient Records – Association Rule Mining**

The given dataset contains **5 patient transactions**. The constraints are:

- **Minimum support = 50%**
- **Minimum confidence = 70%**
- Every generated rule **must include “Fever”**.

#### **1. Given Dataset**

##### **Patient Symptoms**

P1      Fever, Cough  
P2      Fever, Headache  
P3      Cough, Headache  
P4      Fever, Cough, Headache  
P5      Fever, Fatigue

Total transactions = **5**

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#### **2. Support Calculation**

The formula for support is:

$$Support(X) = \frac{\text{Number of transactions containing X}}{\text{Total transactions}} \times 100$$

Minimum support = **50%**

Therefore, an itemset must occur in at least:

$$5 \times 50\% = 2.5$$

Since the number of transactions must be whole, the itemset must occur in **at least 3 transactions**.

##### **Frequent 1-itemsets**

| <b>Item</b> | <b>Occurrences</b> | <b>Support</b> |
|-------------|--------------------|----------------|
| Fever       | 4                  | 80%            |

| Item | Occurrences | Support |
|------|-------------|---------|
|------|-------------|---------|

|       |   |     |
|-------|---|-----|
| Cough | 3 | 60% |
|-------|---|-----|

|          |   |     |
|----------|---|-----|
| Headache | 3 | 60% |
|----------|---|-----|

|         |   |     |
|---------|---|-----|
| Fatigue | 1 | 20% |
|---------|---|-----|

Therefore:

$L_1 = \{\text{Fever, Cough, Headache}\}$

Fatigue is eliminated because its support is only **20%**, which is below 50%.

### 3. Frequent 2-itemsets

Now consider itemsets containing **Fever**, because the constraint requires every rule to include Fever.

**Fever + Cough**

Occurs in:

- P1
- P4

$$\text{Support}(F, \text{everCough}) = \frac{2}{5} \times 100 = 40\%$$

Since **40% < 50%**, it is **not frequent**.

**Fever + Headache**

Occurs in:

- P2
- P4

$$\text{Support}(F, \text{everHeadache}) = \frac{2}{5} \times 100 = 40\%$$

Since **40% < 50%**, it is **not frequent**.

Therefore, there are **no frequent 2-itemsets containing Fever**.

### 4. Confidence Calculation

The confidence formula is:

$$\text{Confidence}(X \rightarrow Y) = \frac{\text{Support}(X \cup Y)}{\text{Support}(X)} \times 100$$

Consider the possible rules involving Fever.

**Rule 1: Fever  $\rightarrow$  Cough**

$$\begin{aligned} \text{Confidence}(\text{Fever} \rightarrow \text{Cough}) &= \frac{\text{Support}(F, \text{everCough})}{\text{Support}(\text{Fever})} \\ &= \frac{2/5}{4/5} \times 100 = 50\% \end{aligned}$$

Required confidence = 70%.

Therefore:

**Fever  $\rightarrow$  Cough is invalid.**

It also fails the minimum support constraint because its support is only 40%.

**Rule 2: Fever  $\rightarrow$  Headache**

$$\begin{aligned} \text{Confidence}(\text{Fever} \rightarrow \text{Headache}) &= \frac{\text{Support}(F, \text{everHeadache})}{\text{Support}(\text{Fever})} \\ &= \frac{2/5}{4/5} \times 100 = 50\% \end{aligned}$$

Therefore:

**Fever  $\rightarrow$  Headache is invalid.**

It fails both the support and confidence requirements.

**5. Other Possible Rules**

Consider:

**Cough  $\rightarrow$  Fever**

$$\text{Confidence}(\text{Cough} \rightarrow \text{Fever}) = \frac{2/5}{3/5} \times 100 = 66.67\%$$

Since **66.67% < 70%**, this rule is invalid.

Also, its support is only **40%**, which is below 50%.

Similarly:

**Headache  $\rightarrow$  Fever**

$$\text{Confidence}(\text{Headache} \rightarrow \text{Fever}) = \frac{2/5}{3/5} \times 100 = 66.67\%$$

This also fails the **70% confidence** and **50% support** requirements.

**6. Final Valid Association Rules**

| Association Rule | Support | Confidence | Result    |
|------------------|---------|------------|-----------|
| Fever → Cough    | 40%     | 50%        | ✗ Invalid |
| Cough → Fever    | 40%     | 66.67%     | ✗ Invalid |
| Fever → Headache | 40%     | 50%        | ✗ Invalid |
| Headache → Fever | 40%     | 66.67%     | ✗ Invalid |

**Final Answer: No valid association rule exists.**

Although **Fever** itself is frequent with **80% support**, combining it with Cough or Headache gives only **40% support**, which is below the required **50%**. Furthermore, the corresponding confidence values are also below the required **70%**.

## 7. Justification

The constraints eliminate all possible rules:

1. **Minimum support of 50%** requires an itemset to occur in at least 3 of the 5 transactions.
2. **Fever** occurs in 4 transactions, so it is frequent.
3. **Fever + Cough** occurs only twice → **40% support**.
4. **Fever + Headache** occurs only twice → **40% support**.
5. The rules involving Fever have confidence values of either **50% or 66.67%**, both below 70%.
6. Therefore, **no association rule satisfies all three constraints simultaneously**.

**Conclusion:** Under the given constraints of **50% minimum support, 70% minimum confidence, and mandatory inclusion of “Fever,”** there are no valid association rules in the Hospital Patient Records dataset.

## Q2. Online Learning Platform – Frequent Itemset Mining

The dataset contains **5 students and their enrolled courses**. The question requires us to generate frequent itemsets with **minimum support = 40%**, with the constraint that **each itemset must contain at least two courses**.

### 1. Given Dataset

#### Student Courses Enrolled

|    |                       |
|----|-----------------------|
| S1 | Python, SQL           |
| S2 | Python, Data Science  |
| S3 | SQL, Machine Learning |

## Student Courses Enrolled

S4 Python, SQL, Data Science

S5 Python, Machine Learning

Total transactions = **5**

Minimum support = **40%**

Therefore, an itemset must occur in at least:

$$5 \times 40\% = 2$$

So, an itemset must appear in **at least 2 student records** to be frequent.

## 2. Apply the Constraint

The important constraint is:

**Every itemset must contain at least two courses.**

Therefore, **1-itemsets are not considered** as final candidates.

For example:

- {Python} → rejected because it contains only one course.
- {SQL} → rejected because it contains only one course.
- {Python, SQL} → considered because it contains two courses.

This constraint helps reduce unnecessary candidate generation.

## 3. Generate 2-Course Itemsets

The possible 2-course combinations are:

1. Python + SQL
2. Python + Data Science
3. Python + Machine Learning
4. SQL + Data Science
5. SQL + Machine Learning
6. Data Science + Machine Learning

Now calculate their support.

**Itemset 1: {Python, SQL}**

Appears in:

- S1

- S4

Number of occurrences = 2

$$Support = \frac{2}{5} \times 100 = 40\%$$

Since **40% ≥ 40%**, it is **frequent**.

### Itemset 2: {Python, Data Science}

Appears in:

- S2
- S4

Number of occurrences = 2

$$Support = \frac{2}{5} \times 100 = 40\%$$

Therefore, it is **frequent**.

### Itemset 3: {Python, Machine Learning}

Appears in:

- S5 only

Number of occurrences = 1

$$Support = \frac{1}{5} \times 100 = 20\%$$

Since **20% < 40%**, it is **not frequent**.

### Itemset 4: {SQL, Data Science}

Appears in:

- S4 only

Number of occurrences = 1

$$Support = \frac{1}{5} \times 100 = 20\%$$

Therefore, it is **not frequent**.

### Itemset 5: {SQL, Machine Learning}

Appears in:

- S3 only

Number of occurrences = 1

$$Support = \frac{1}{5} \times 100 = 20\%$$

Therefore, it is **not frequent**.

#### Itemset 6: {Data Science, Machine Learning}

No student has both courses.

Number of occurrences = 0

$$Support = \frac{0}{5} \times 100 = 0\%$$

Therefore, it is **not frequent**.

#### 4. Frequent 2-Itemsets

| Itemset                          | Occurrences | Support | Status         |
|----------------------------------|-------------|---------|----------------|
| {Python, SQL}                    | 2           | 40%     | ✓ Frequent     |
| {Python, Data Science}           | 2           | 40%     | ✓ Frequent     |
| {Python, Machine Learning}       | 1           | 20%     | ✗ Not frequent |
| {SQL, Data Science}              | 1           | 20%     | ✗ Not frequent |
| {SQL, Machine Learning}          | 1           | 20%     | ✗ Not frequent |
| {Data Science, Machine Learning} | 0           | 0%      | ✗ Not frequent |

**Therefore:**

$$L_2 = \{\{P, ythonSQL\}, \{P, ythonData Science\}\}$$

#### 5. Check 3-Course Itemsets

The possible 3-course combinations include:

- {Python, SQL, Data Science}
- {Python, SQL, Machine Learning}

- {Python, Data Science, Machine Learning}
- {SQL, Data Science, Machine Learning}

**{Python, SQL, Data Science}**

Appears only in S4.

$$Support = \frac{1}{5} \times 100 = 20\%$$

So it is **not frequent**.

The other 3-course combinations occur in fewer than 2 transactions and therefore also fail the 40% support requirement.

Thus, **there are no frequent 3-course itemsets**.

## 6. Final Frequent Itemsets

The frequent itemsets satisfying **both constraints** are:

### Frequent Itemset 1

$$\{P, ythonSQL\}$$

Support = **40%**

### Frequent Itemset 2

$$\{P, ythonData Science\}$$

Support = **40%**

There are **no frequent itemsets containing three or more courses**.

## 7. How the Constraint Helps in Pruning

The constraint that **each itemset must contain at least two courses** helps the Apriori/frequent-itemset mining process by eliminating all single-course itemsets from the required search space.

For example, without the constraint, the algorithm would first consider:

- {Python}
- {SQL}
- {Data Science}
- {Machine Learning}

and then generate larger combinations.

With the constraint, the mining process can directly focus on combinations such as:



- {Python, SQL}
- {Python, Data Science}
- {Python, Machine Learning}

This reduces:

- **Number of candidates**
- **Support calculations**
- **Database scanning**
- **Memory usage**
- **Overall computational effort**

The constraint therefore acts as a **pruning condition**, allowing unqualified candidates to be discarded early.

## 8. Conclusion

With a **minimum support of 40%** and the constraint that every itemset must contain **at least two courses**, the valid frequent itemsets are:

$$\{P, ythonSQL\} \text{ Support} = 40\%$$

$$\{P, ythonData \text{ Science}\} \text{ Support} = 40\%$$

The constraint improves mining efficiency by **pruning single-course itemsets and reducing the number of candidate combinations that need to be evaluated**. This makes the association-rule mining process faster and more efficient.

## Q3. Library Borrowing Records – Apriori Algorithm

The question asks us to apply the **Apriori algorithm** to the given library borrowing dataset, with the constraint that the **maximum itemset size is 3**. We must identify the frequent itemsets and explain how this constraint reduces computational complexity during candidate generation.

**Note:** The question statement specifies the maximum itemset size but does **not explicitly provide a minimum support threshold**. Therefore, based strictly on the supplied question, we can identify the itemsets and their support counts, but whether an itemset is called "frequent" depends on the minimum-support value chosen by the assessment/instructor.

### 1. Given Dataset

#### Transaction Books Borrowed

|    |                                      |
|----|--------------------------------------|
| T1 | Data Mining, Python Programming      |
| T2 | Data Mining, Database Systems        |
| T3 | Python Programming, Database Systems |

### Transaction Books Borrowed

T4            Data Mining, Python Programming, Database Systems

T5            Data Mining, Artificial Intelligence

Total transactions:

$$N = 5$$

Items:

- Data Mining
- Python Programming
- Database Systems
- Artificial Intelligence

### 2. Apriori Algorithm

The **Apriori algorithm** generates frequent itemsets level by level.

The basic steps are:

1. Generate candidate 1-itemsets.
2. Calculate their support.
3. Remove infrequent itemsets.
4. Generate candidate 2-itemsets.
5. Calculate their support.
6. Generate candidate 3-itemsets.
7. Stop after size 3 because of the given constraint.

The maximum itemset size is:

$$\boxed{3}$$

Therefore, **4-itemsets and larger itemsets are never generated.**

### 3. Candidate 1-Itemsets

Count the occurrence of each book.

| Itemset              | Occurrences | Support |
|----------------------|-------------|---------|
| {Data Mining}        | 4           | 80%     |
| {Python Programming} | 2           | 40%     |

| Itemset | Occurrences | Support |
|---------|-------------|---------|
|---------|-------------|---------|

|                    |   |     |
|--------------------|---|-----|
| {Database Systems} | 3 | 60% |
|--------------------|---|-----|

|                           |   |     |
|---------------------------|---|-----|
| {Artificial Intelligence} | 1 | 20% |
|---------------------------|---|-----|

Support is calculated as:

$$Support(X) = \frac{\text{Number of transactions containing } X}{5} \times 100$$

Thus:

$$Support(Data Mining) = \frac{4}{5} \times 100 = 80\%$$

$$Support(Python Programming) = \frac{2}{5} \times 100 = 40\%$$

$$Support(Database Systems) = \frac{3}{5} \times 100 = 60\%$$

$$Support(Artificial Intelligence) = \frac{1}{5} \times 100 = 20\%$$

#### 4. Candidate 2-Itemsets

There are six possible pairs.

##### 1. {Data Mining, Python Programming}

Appears in:

- T1
- T4

Therefore:

$$Support = \frac{2}{5} \times 100 = 40\%$$

##### 2. {Data Mining, Database Systems}

Appears in:

- T2
- T4

Therefore:

$$Support = \frac{2}{5} \times 100 = 40\%$$

### 3. {Data Mining, Artificial Intelligence}

Appears in:

- T5

$$Support = \frac{1}{5} \times 100 = 20\%$$

### 4. {Python Programming, Database Systems}

Appears in:

- T3
- T4

$$Support = \frac{2}{5} \times 100 = 40\%$$

### 5. {Python Programming, Artificial Intelligence}

Appears in no transaction.

$$Support = 0\%$$

### 6. {Database Systems, Artificial Intelligence}

Appears in no transaction.

$$Support = 0\%$$

### 2-Itemset Support Table

| 2-Itemset                                     | Count | Support |
|-----------------------------------------------|-------|---------|
| {Data Mining, Python Programming}             | 2     | 40%     |
| {Data Mining, Database Systems}               | 2     | 40%     |
| {Data Mining, Artificial Intelligence}        | 1     | 20%     |
| {Python Programming, Database Systems}        | 2     | 40%     |
| {Python Programming, Artificial Intelligence} | 0     | 0%      |
| {Database Systems, Artificial Intelligence}   | 0     | 0%      |

### 5. Candidate 3-Itemsets

The possible 3-itemsets are:

**{Data Mining, Python Programming, Database Systems}**

Appears in:

- T4

Therefore:

$$Support = \frac{1}{5} \times 100 = 20\%$$

**{Data Mining, Python Programming, Artificial Intelligence}**

No transaction contains all three.

$$Support = 0\%$$

**{Data Mining, Database Systems, Artificial Intelligence}**

No transaction contains all three.

$$Support = 0\%$$

**{Python Programming, Database Systems, Artificial Intelligence}**

No transaction contains all three.

$$Support = 0\%$$

### 3-Itemset Table

| 3-Itemset                                                       | Count | Support |
|-----------------------------------------------------------------|-------|---------|
| {Data Mining, Python Programming, Database Systems}             | 1     | 20%     |
| {Data Mining, Python Programming, Artificial Intelligence}      | 0     | 0%      |
| {Data Mining, Database Systems, Artificial Intelligence}        | 0     | 0%      |
| {Python Programming, Database Systems, Artificial Intelligence} | 0     | 0%      |

## 6. Frequent Itemsets

Since the assessment does **not state a minimum-support value**, the exact final set of "frequent" itemsets cannot be determined from the question alone.

However, the complete support information is:

### 1-itemsets

- {Data Mining} → **80%**
- {Python Programming} → **40%**
- {Database Systems} → **60%**

- {Artificial Intelligence} → **20%**

## 2-itemsets

- {Data Mining, Python Programming} → **40%**
- {Data Mining, Database Systems} → **40%**
- {Python Programming, Database Systems} → **40%**
- {Data Mining, Artificial Intelligence} → **20%**
- {Python Programming, Artificial Intelligence} → **0%**
- {Database Systems, Artificial Intelligence} → **0%**

## 3-itemsets

- {Data Mining, Python Programming, Database Systems} → **20%**
- Other 3-itemsets → **0%**

For example, **if the intended minimum support is 40%**, the frequent itemsets would be:

$$\boxed{\{Data Mining\}, \{Python Programming\}, \{Database Systems\}}$$

and

$$\boxed{\{Data Mining, Python Programming\}, \{Data Mining, Database Systems\}, \{Python Programming, Database Systems\}}$$

The 3-itemset has only 20% support and would not be frequent under a 40% threshold.

## 7. How Maximum Size 3 Reduces Computational Complexity

The constraint is:

$$\boxed{Maximum\ itemset\ size = 3}$$

Without this constraint, Apriori could continue generating:

$$1\text{-itemsets} \rightarrow 2\text{-itemsets} \rightarrow 3\text{-itemsets} \rightarrow 4\text{-itemsets} \rightarrow \dots$$

But here, the algorithm stops after generating 3-itemsets.

For the **4 distinct books** in this dataset, the possible combinations are:

$$\begin{aligned} \binom{4}{1} &= 4 \\ \binom{4}{2} &= 6 \\ \binom{4}{3} &= 4 \end{aligned}$$

A 4-itemset would require:

$$\binom{4}{4} = 1$$

But because the maximum size is 3, this final candidate is **never generated**.

Thus, instead of considering all possible combinations up to size 4, Apriori only considers:

$$4 + 6 + 4 = 14$$

possible itemsets.

This reduces:

- Candidate generation
- Support counting
- Database scans
- Memory requirements
- Computational time

## 8. Final Conclusion

The Apriori algorithm identifies itemsets level by level, starting from individual books and generating larger combinations. The given **maximum itemset size of 3** prevents the algorithm from generating any 4-itemsets or larger combinations. This significantly reduces the candidate search space and computational effort.

**Important:** Since the supplied Q3 does not specify a minimum-support percentage, the support values above should be used to determine the final frequent itemsets once the instructor's intended threshold is known.

## Q4. University Course Hierarchy – Multilevel Association Rule Mining

The dataset contains students' courses, and the question asks us to apply **multilevel association rule mining** with the constraint that rules should be generated **only at the “Computer Science” level**. We also need to explain how the course hierarchy affects the discovery of association rules.

### 1. Given Dataset

#### Student Courses Taken

|    |                                                 |
|----|-------------------------------------------------|
| S1 | Engineering, Computer Science, Machine Learning |
| S2 | Engineering, Information Technology             |
| S3 | Engineering, Computer Science                   |
| S4 | Engineering, Computer Science, Machine Learning |

## Student Courses Taken

S5      Engineering, Information Technology

Total students/transactions:

$$N = 5$$

## 2. What is Multilevel Association Rule Mining?

**Multilevel association rule mining** discovers relationships among items at different levels of a hierarchy.

For example, courses can be organized as:

Engineering

|

|— Computer Science

|    |— Machine Learning

|

|— Information Technology

In this question, the required constraint is:

Generate rules only at the Computer Science level

Therefore, we focus on the association involving **Computer Science** and do not generate rules specifically at the Information Technology level.

## 3. Identify Computer Science Transactions

From the dataset:

- S1 → Computer Science
- S2 → Information Technology
- S3 → Computer Science
- S4 → Computer Science
- S5 → Information Technology

Therefore, Computer Science occurs in:

S1, S3, S4

Number of occurrences:



Support of Computer Science:

$$\text{Support}(\text{Computer Science}) = \frac{3}{5} \times 100$$

$$\boxed{\text{Support} = 60\%}$$

Thus, **Computer Science appears in 60% of the student records.**

#### 4. Association with Machine Learning

Now consider:

$$\{C, \text{omputer ScienceMachine Learning}\}$$

This combination occurs in:

- S1
- S4

Therefore:

$$\text{Support}(C, SML) = \frac{2}{5} \times 100$$

$$\boxed{\text{Support}(C, SML) = 40\%}$$

So, **Computer Science and Machine Learning occur together in 40% of the records.**

#### 5. Generate Association Rules

The main possible rule at the required level is:

**Rule 1: Computer Science  $\rightarrow$  Machine Learning**

Confidence is calculated as:

$$\text{Confidence}(CS \rightarrow ML) = \frac{\text{Support}(C, SML)}{\text{Support}(CS)} \times 100$$

Substituting:

$$= \frac{40\%}{60\%} \times 100$$

$$\boxed{\text{Confidence} = 66.67\%}$$

Therefore:

**Computer Science  $\rightarrow$  Machine Learning**

has:

- **Support = 40%**
- **Confidence = 66.67%**

### Reverse Rule: Machine Learning → Computer Science

Machine Learning occurs in:

- S1
- S4

Therefore:

$$\text{Support}(\text{Machine Learning}) = \frac{2}{5} \times 100 = 40\%$$

Confidence:

$$\text{Confidence}(ML \rightarrow CS) = \frac{40\%}{40\%} \times 100$$

$\text{Confidence} = 100\%$

Therefore:

### Machine Learning → Computer Science

has:

- **Support = 40%**
- **Confidence = 100%**

This means that **every student in this dataset who took Machine Learning also took Computer Science.**

## 6. Summary of Rules

| Association Rule                    | Support | Confidence |
|-------------------------------------|---------|------------|
| Computer Science → Machine Learning | 40%     | 66.67%     |
| Machine Learning → Computer Science | 40%     | 100%       |

The question does not provide a minimum support or confidence threshold, so these rules are reported based on their calculated values rather than declaring one definitively "valid" or "invalid."

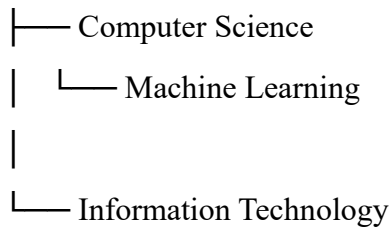
## 7. Effect of the Course Hierarchy

The course hierarchy helps organize courses from **general to specific levels.**

For example:

Engineering

|



If association mining were performed at all levels, the algorithm could generate rules involving:

- Engineering
- Computer Science
- Information Technology
- Machine Learning

However, the question specifically imposes the constraint:

Rules only at Computer Science level

Therefore, the search is restricted to relationships involving **Computer Science**.

**Benefits of this constraint:**

1. **Reduces search space** – unnecessary rules from other branches are ignored.
2. **Improves efficiency** – fewer candidate itemsets need to be examined.
3. **Produces focused results** – the discovered patterns are specifically related to Computer Science.
4. **Avoids irrelevant associations** – relationships involving the Information Technology branch are excluded.
5. **Supports targeted analysis** – universities can focus on course combinations relevant to Computer Science students.

## 8. Interpretation of the Pattern

The important pattern is:

*Machine Learning* → *Computer Science*

with **100% confidence** in this dataset.

This indicates that all students who selected **Machine Learning** also selected **Computer Science**.

The reverse relationship:

*Computer Science* → *Machine Learning*

has only **66.67% confidence**, because 3 students took Computer Science but only 2 of them also took Machine Learning.

Thus, Machine Learning has a strong association with Computer Science in this particular dataset.

## 9. Final Conclusion

Multilevel association rule mining uses the course hierarchy to discover relationships at a specified level. Under the given constraint, the analysis is restricted to the **Computer Science level**, which reduces the number of candidate rules and makes the mining process more focused.

The key discovered relationship is:

|                                                   |
|---------------------------------------------------|
| <i>Machine Learning</i> → <i>Computer Science</i> |
|---------------------------------------------------|

with:

|                                                |
|------------------------------------------------|
| <i>Support</i> = 40%, <i>Confidence</i> = 100% |
|------------------------------------------------|

This shows that **Machine Learning is strongly associated with Computer Science** in the given student records.

## Q5. Insurance Customer Records – Multidimensional Association Rule Mining

The question asks us to apply **multidimensional association rule mining** with the constraint **Age Group = “Young”**. We need to identify meaningful patterns involving **income level and policy type** and explain how these patterns can support targeted marketing.

### 1. Given Dataset

#### Customer Age Group Income Level Policy Purchased

|    |        |        |                   |
|----|--------|--------|-------------------|
| C1 | Young  | High   | Health Insurance  |
| C2 | Middle | Medium | Vehicle Insurance |
| C3 | Young  | Low    | Travel Insurance  |
| C4 | Senior | High   | Health Insurance  |
| C5 | Young  | Medium | Vehicle Insurance |

Total customers = 5

### 2. Apply the Constraint

The given constraint is:

$$\boxed{Age\ Group = Young}$$

Therefore, we remove customers who belong to the **Middle** and **Senior** age groups.

### Filtered Dataset

#### Customer Age Group Income Level Policy

|    |       |        |                   |
|----|-------|--------|-------------------|
| C1 | Young | High   | Health Insurance  |
| C3 | Young | Low    | Travel Insurance  |
| C5 | Young | Medium | Vehicle Insurance |

Number of Young customers:

$$\boxed{3}$$

So, the association analysis is performed only on **C1, C3 and C5**.

### 3. Identify Income-Level Patterns

Among the Young customers:

- High income  $\rightarrow$  1 customer
- Medium income  $\rightarrow$  1 customer
- Low income  $\rightarrow$  1 customer

Therefore:

$$Support(Y, oungeHigh) = \frac{1}{3} \times 100 = \boxed{33.33\%}$$

$$Support(Y, oungeMedium) = \frac{1}{3} \times 100 = \boxed{33.33\%}$$

$$Support(Y, oungeLow) = \frac{1}{3} \times 100 = \boxed{33.33\%}$$

Thus, the three income categories are **equally represented** among the Young customers.

### 4. Identify Policy Patterns

Among the Young customers:

- High income  $\rightarrow$  **Health Insurance**
- Low income  $\rightarrow$  **Travel Insurance**
- Medium income  $\rightarrow$  **Vehicle Insurance**

Therefore, each income level corresponds to a different policy type in this dataset.

| Income Level Policy |                   | Customers Support among Young |        |
|---------------------|-------------------|-------------------------------|--------|
| High                | Health Insurance  | 1                             | 33.33% |
| Low                 | Travel Insurance  | 1                             | 33.33% |
| Medium              | Vehicle Insurance | 1                             | 33.33% |

## 5. Association Rules

Because the question focuses on **Age Group, Income Level and Policy Type**, we can express the observed patterns as multidimensional rules.

### Rule 1

$$Young \wedge High\ Income \rightarrow Health\ Insurance$$

Support among all customers:

$$\frac{1}{5} \times 100 = \boxed{20\%}$$

Confidence:

$$\begin{aligned} & \frac{Young + High + Health}{Young + High} \times 100 \\ &= \frac{1}{1} \times 100 = \boxed{100\%} \end{aligned}$$

So, within this dataset, every Young customer with High income purchased Health Insurance.

### Rule 2

$$Young \wedge Low\ Income \rightarrow Travel\ Insurance$$

Support:

$$\frac{1}{5} \times 100 = \boxed{20\%}$$

Confidence:

$$\frac{1}{1} \times 100 = \boxed{100\%}$$

Thus, every Young customer with Low income purchased Travel Insurance.

### Rule 3

$$Young \wedge Medium\ Income \rightarrow Vehicle\ Insurance$$

Support:

$$\frac{1}{5} \times 100 = \boxed{20\%}$$

Confidence:

$$\frac{1}{1} \times 100 = \boxed{100\%}$$

Thus, every Young customer with Medium income purchased Vehicle Insurance.

## 6. Summary of Meaningful Patterns

| Rule                                                  | Support Confidence |      |
|-------------------------------------------------------|--------------------|------|
| Young + High Income $\rightarrow$ Health Insurance    | 20%                | 100% |
| Young + Low Income $\rightarrow$ Travel Insurance     | 20%                | 100% |
| Young + Medium Income $\rightarrow$ Vehicle Insurance | 20%                | 100% |

The **100% confidence** occurs because each income group has only one Young customer in this very small dataset.

## 7. How These Patterns Support Targeted Marketing

These patterns can help an insurance company create different marketing strategies for Young customers.

### 1. Young + High Income $\rightarrow$ Health Insurance

High-income young customers can be targeted with:

- Premium health insurance plans
- Comprehensive health coverage
- Additional health benefits
- Higher-value insurance packages

### 2. Young + Low Income $\rightarrow$ Travel Insurance

Low-income young customers can be targeted with:

- Affordable travel insurance
- Low-cost policies
- Short-term travel coverage

- Budget-friendly packages

### 3. Young + Medium Income → Vehicle Insurance

Medium-income young customers can be targeted with:

- Vehicle insurance packages
- Affordable vehicle coverage
- Customized vehicle protection plans
- Discounts for young vehicle owners

## 8. Importance of Multidimensional Association Mining

Traditional association mining may focus mainly on relationships between items.

**Multidimensional association rule mining** considers multiple attributes simultaneously.

Here, three dimensions are considered:

$$\text{Age Group} + \text{Income Level} + \text{Policy Type}$$

The constraint:

$$\text{Age Group} = \text{Young}$$

allows the company to focus specifically on Young customers.

This makes the analysis more useful for **customer segmentation and targeted marketing**.

## 9. Final Conclusion

After applying the constraint **Age Group = Young**, three customers remain: C1, C3 and C5.

The observed patterns are:

$$\text{Young} + \text{High Income} \rightarrow \text{Health Insurance}$$

$$\text{Young} + \text{Low Income} \rightarrow \text{Travel Insurance}$$

$$\text{Young} + \text{Medium Income} \rightarrow \text{Vehicle Insurance}$$

Each rule has **20% support among the complete dataset and 100% confidence** in this small sample.

Therefore, the insurance company can use these patterns to **segment Young customers according to income level and promote suitable policy types**, improving the effectiveness of targeted marketing.